

Trigonometry

ANSWER KEY



Right-triangle trigonometry

- 1 In a right triangle, the angle θ has an opposite side of 7 and hypotenuse 25. Find $\cos \theta$ and $\tan \theta$.
Adjacent side: $\sqrt{25^2 - 7^2} = \sqrt{625 - 49} = \sqrt{576} = 24$. $\cos \theta = \frac{24}{25}$. $\tan \theta = \frac{7}{24}$.
- 2 In a right triangle, angle θ has an adjacent side of 8 and opposite side 15. Find $\sin \theta$ and the hypotenuse.
Hypotenuse: $\sqrt{8^2 + 15^2} = \sqrt{64 + 225} = \sqrt{289} = 17$. $\sin \theta = \frac{15}{17}$.

Exact values and the unit circle

- 3 State the exact value of $\sin\left(\frac{\pi}{3}\right)$ and $\cos(45^\circ)$.
 $\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$. $\cos(45^\circ) = \frac{\sqrt{2}}{2}$.
- 4 State the exact value of $\tan(60^\circ)$ and $\sin(30^\circ)$.
 $\tan(60^\circ) = \sqrt{3}$. $\sin(30^\circ) = \frac{1}{2}$.

Graphs of trigonometric functions

- 5 State the amplitude and period of $y = 4\sin(2x)$.
Amplitude: 4. Period: $\frac{2\pi}{2} = \pi$.
- 6 State the amplitude and period of $y = 3\cos\left(\frac{x}{2}\right)$.
Amplitude: 3. Period: $\frac{2\pi}{1/2} = 4\pi$.

Trigonometric identities

- 7 Given $\sin \theta = \frac{3}{5}$ with θ acute, find $\cos \theta$ using the identity $\sin^2 \theta + \cos^2 \theta = 1$.
 $\cos^2 \theta = 1 - \frac{9}{25} = \frac{16}{25}$. Since θ is acute, $\cos \theta = \frac{4}{5}$.
- 8 Simplify $\frac{\sin \theta}{1 - \cos \theta} \times (1 - \cos \theta)$.
The $(1 - \cos \theta)$ factors cancel, leaving $\sin \theta$.
- 9 Given $\cos \theta = \frac{5}{13}$ with θ acute, find $\sin \theta$ using the identity $\sin^2 \theta + \cos^2 \theta = 1$.
 $\sin^2 \theta = 1 - \frac{25}{169} = \frac{144}{169}$. Since θ is acute, $\sin \theta = \frac{12}{13}$.

Solving trigonometric equations

- 10 Solve $2\sin\theta = 1$ for $0^\circ \leq \theta < 360^\circ$.
 $\sin\theta = \frac{1}{2}$. Solutions: $\theta = 30^\circ$ or $\theta = 150^\circ$.
- 11 Solve $\cos(2\theta) = -1$ for $0^\circ \leq \theta < 360^\circ$.
 $2\theta = 180^\circ$ or $2\theta = 540^\circ$ (within $0^\circ \leq 2\theta < 720^\circ$), so $\theta = 90^\circ$ or $\theta = 270^\circ$.
- 12 Solve $\tan\theta = -1$ for $0^\circ \leq \theta < 360^\circ$.
 $\theta = 135^\circ$ or $\theta = 315^\circ$.

Law of Sines and Law of Cosines

- 13 In triangle ABC , $a = 9$, $b = 12$, and angle $C = 55^\circ$. Find side c , to the nearest tenth, using the Law of Cosines.
 $c^2 = 9^2 + 12^2 - 2(9)(12)\cos(55^\circ) = 81 + 144 - 216(0.5736) \approx 225 - 123.9 \approx 101.1$.
 $c \approx \sqrt{101.1} \approx 10.1$.
- 14 In triangle ABC , angle $A = 40^\circ$, angle $B = 70^\circ$, and $a = 8$. Find side b , to the nearest tenth, using the Law of Sines.
 $\frac{a}{\sin A} = \frac{b}{\sin B}$: $\frac{8}{\sin 40^\circ} = \frac{b}{\sin 70^\circ}$. $b = \frac{8\sin 70^\circ}{\sin 40^\circ} \approx \frac{8(0.9397)}{0.6428} \approx 11.7$.
- 15 In triangle ABC , $a = 7$, $b = 10$, $c = 13$. Find the largest angle, to the nearest degree, using the Law of Cosines.
The largest angle is opposite the longest side, so we find angle C :
 $\cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{49 + 100 - 169}{2(7)(10)} = \frac{-20}{140} \approx -0.1429$. $C = \cos^{-1}(-0.1429) \approx 98^\circ$.

Trigonometric identities for compound angles

- 16 Use the identity $\sin(A + B) = \sin A \cos B + \cos A \sin B$ to find the exact value of $\sin(75^\circ)$, using $75^\circ = 45^\circ + 30^\circ$.
 $\sin(75^\circ) = \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \cdot \frac{1}{2} = \frac{\sqrt{6} + \sqrt{2}}{4}$.
- 17 Use the identity $\cos(A - B) = \cos A \cos B + \sin A \sin B$ to find the exact value of $\cos(15^\circ)$, using $15^\circ = 45^\circ - 30^\circ$.
 $\cos(15^\circ) = \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \cdot \frac{1}{2} = \frac{\sqrt{6} + \sqrt{2}}{4}$.

A well-designed word problem: the height of a lighthouse

- 18** This question has two parts. A ship observes a lighthouse at an angle of elevation of 18° . After sailing 500 meters directly toward the lighthouse, the angle of elevation increases to 30° . (a) Let h be the lighthouse's height and d the final distance from its base. Write two equations relating h and d using the tangent ratio. (b) Solve the system to find h , to the nearest meter.

(a) At the closer position: $\tan(30^\circ) = \frac{h}{d}$, so $h = d \tan(30^\circ)$. At the farther position (distance $d + 500$): $\tan(18^\circ) = \frac{h}{d + 500}$, so $h = (d + 500) \tan(18^\circ)$. (b) Setting equal: $d \tan(30^\circ) = (d + 500) \tan(18^\circ)$. Using $\tan(30^\circ) \approx 0.5774$ and $\tan(18^\circ) \approx 0.3249$: $0.5774d = 0.3249d + 162.45$, so $0.2525d = 162.45$, giving $d \approx 643.4$. Then $h = 643.4 \times 0.5774 \approx 372$ meters.